UNIVERSITÀ DEGLI STUDI DI PISA

CORSO DI STUDIO IN BIONICS ENGINEERING

STATISTICAL SIGNAL PROCESSING - *Prova scritta preliminare del 07/06/2018*

PROBLEM - We observe the following two samples: $X_1 = A_1 \cdot \sqrt{\theta} + W_1$ and $X_2 = A_2 \cdot \sqrt{\theta} + W_2$, where θ is a positive (θ>0) deterministic unknown parameter that we want to estimate. Parameters A_1, A_2, W_1, W_2 can be modelled as mutually independent Gaussian random variables distributed as follows: $A_1, A_2 \in N\left(0, \sigma_A^2\right), \quad W_1, W_2 \in N\left(0, \sigma_W^2\right)$.

(1) Derive the joint probability density function (pdf) of the two observed data samples $\mathbf{X} = \left[X_1 \quad X_2\right]^T$ and the log-likelihood function ln$L(\theta)$.

(2) Derive the Maximum Likelihood (ML) estimator of θ given the observation of **X**.

(3) Derive the bias of $\hat{\theta}_{ML}$.

(4) Derive the mean square error (MSE) of $\hat{\theta}_{ML}$ and the Cramér-Rao lower bound (CRB) on the estimation accuracy of parameter θ and establish if the estimator is efficient. Express the CRB as a function of $\gamma \triangleq \dfrac{\sigma_A^2}{\sigma_W^2}$.

Solution

(1) Derive the joint pdf of the two observed data samples $\mathbf{X} = \left[X_1 \quad X_2 \right]^T$; derive and plot the log-likelihood function $\ln L(\theta)$.

X_1 and X_2 are jointly Gaussian random variables, hence the pdf is specified by the 5 parameters:

$$\eta_1 \triangleq E\{X_1\} = E\left\{A_1 \cdot \sqrt{\theta} + W_1\right\} = E\{A_1\}\sqrt{\theta} + E\{W_1\} = 0$$

$$\eta_2 \triangleq E\{X_2\} = E\left\{A_2 \cdot \sqrt{\theta} + W_2\right\} = E\{A_2\}\sqrt{\theta} + E\{W_2\} = 0$$

$$\sigma_1^2 \triangleq \mathrm{var}\{X_1\} = E\left\{\left(A_1 \cdot \sqrt{\theta} + W_1\right)^2\right\} = \sigma_A^2\theta + \sigma_W^2$$

$$\sigma_2^2 \triangleq \mathrm{var}\{X_2\} = E\left\{\left(A_2 \cdot \sqrt{\theta} + W_2\right)^2\right\} = \sigma_A^2\theta + \sigma_W^2$$

$$\rho = \frac{\mathrm{cov}\{X_1, X_2\}}{\sigma_A^2\theta + \sigma_W^2} = \frac{E\{X_1 X_2\}}{\sigma_A^2\theta + \sigma_W^2} = \frac{E\left\{\left(A_1 \cdot \sqrt{\theta} + W_1\right)\left(A_2 \cdot \sqrt{\theta} + W_2\right)\right\}}{\sigma_A^2\theta + \sigma_W^2} = \frac{E\left\{A_1 \cdot \sqrt{\theta} + W_1\right\} E\left\{A_2 \cdot \sqrt{\theta} + W_2\right\}}{\sigma_A^2\theta + \sigma_W^2} = 0$$

Since the two Gaussian r.v. are uncorrelated they are also independent:

$$f_{X_1X_2}(x_1, x_2) = f_{X_1}(x_1) f_{X_2}(x_2) = \frac{1}{\sqrt{2\pi\left(\sigma_A^2\theta + \sigma_W^2\right)}} e^{-\frac{x_1^2}{2\left(\sigma_A^2\theta + \sigma_W^2\right)}} \frac{1}{\sqrt{2\pi\left(\sigma_A^2\theta + \sigma_W^2\right)}} e^{-\frac{x_2^2}{2\left(\sigma_A^2\theta + \sigma_W^2\right)}}$$

$$= \frac{1}{2\pi\left(\sigma_A^2\theta + \sigma_W^2\right)} e^{-\frac{x_1^2 + x_2^2}{2\left(\sigma_A^2\theta + \sigma_W^2\right)}}$$

$$\ln L(\theta) = \ln f_{X_1X_2}(x_1, x_2) = -\ln(2\pi) - \ln\left(\sigma_A^2\theta + \sigma_W^2\right) - \frac{x_1^2 + x_2^2}{2\left(\sigma_A^2\theta + \sigma_W^2\right)}$$

(2) Derive the Maximum Likelihood (ML) estimator of θ given the observation of **X**.

$$\frac{\partial \ln L(\theta)}{\partial \theta} = -\frac{\sigma_A^2}{\sigma_A^2\theta+\sigma_W^2} + \frac{\sigma_A^2\left(x_1^2+x_2^2\right)}{2\left(\sigma_A^2\theta+\sigma_W^2\right)^2} = 0 \quad \Rightarrow \quad -1+\frac{\left(x_1^2+x_2^2\right)}{2\left(\sigma_A^2\theta+\sigma_W^2\right)} = 0$$

$$\hat{\theta}_{ML} = \frac{1}{\sigma_A^2}\left(\frac{x_1^2+x_2^2}{2} - \sigma_W^2\right)$$

(3) Derive the bias of $\hat{\theta}_{ML}$.

$$E\left\{\hat{\theta}_{ML}\right\} = E\left\{\frac{1}{\sigma_A^2}\left(\frac{x_1^2+x_2^2}{2}-\sigma_W^2\right)\right\} = \frac{1}{\sigma_A^2}\left(\frac{E\left\{x_1^2+x_2^2\right\}}{2}-\sigma_W^2\right) = \frac{1}{\sigma_A^2}\left(\frac{2\left(\sigma_A^2\theta+\sigma_W^2\right)}{2}-\sigma_W^2\right) = \theta$$

so the estimator is unbiased.

(3) Derive the mean square error (MSE) of $\hat{\theta}_{ML}$ and the Cramér-Rao lower bound (CRB) on the estimation accuracy of parameter θ and establish if the estimator is efficient. Express the CRB as a function of $\gamma \triangleq \frac{\sigma_A^2}{\sigma_W^2}$.

$$\begin{aligned}
\frac{\partial \ln L(\theta)}{\partial \theta} &= -\frac{\sigma_A^2}{\sigma_A^2\theta+\sigma_W^2} + \frac{\sigma_A^2\left(x_1^2+x_2^2\right)}{2\left(\sigma_A^2\theta+\sigma_W^2\right)^2} = \frac{\sigma_A^2}{\left(\sigma_A^2\theta+\sigma_W^2\right)^2}\left[\frac{\left(x_1^2+x_2^2\right)}{2} - \left(\sigma_A^2\theta+\sigma_W^2\right)\right] \\
&= \frac{\sigma_A^4}{\left(\sigma_A^2\theta+\sigma_W^2\right)^2}\left[\frac{1}{\sigma_A^2}\left(\frac{x_1^2+x_2^2}{2}-\sigma_W^2\right)-\theta\right] \\
&= I(\theta)\left[g(\mathbf{X})-\theta\right]
\end{aligned}$$

where $I(\theta) \triangleq \frac{\sigma_A^4}{\left(\sigma_A^2\theta+\sigma_W^2\right)^2} = \left(\frac{\sigma_A^2}{\sigma_A^2\theta+\sigma_W^2}\right)^2, \qquad g(\mathbf{X}) \triangleq \frac{1}{\sigma_A^2}\left(\frac{x_1^2+x_2^2}{2}-\sigma_W^2\right)$

Hence $\hat{\theta}_{ML} = g(\mathbf{X}) = \frac{1}{\sigma_A^2}\left(\frac{x_1^2 + x_2^2}{2} - \sigma_W^2\right)$ is an unbiased and efficient estimator of θ and its MSE coincides with the CRB:

$$MSE\left\{\hat{\theta}_{ML}\right\} = CRB(\theta) = I^{-1}(\theta) = \left(\frac{\sigma_A^2\theta + \sigma_W^2}{\sigma_A^2}\right)^2 = \left(\theta + \frac{1}{\gamma}\right)^2.$$

The MSE can also be derived as follows:

$$MSE\left\{\hat{\theta}_{ML}\right\} = \operatorname{var}\left\{\hat{\theta}_{ML}\right\} = \operatorname{var}\left\{\frac{1}{\sigma_A^2}\left(\frac{X_1^2 + x_2^2}{2} - \sigma_W^2\right)\right\} = \frac{1}{4\sigma_A^4}\operatorname{var}\left\{X_1^2 + X_2^2\right\} = \frac{1}{2\sigma_A^4}\operatorname{var}\left\{X_1^2\right\}$$
$$= \frac{E\left\{X_1^4\right\} - \left(E\left\{X_1^2\right\}\right)^2}{2\sigma_A^4} = \frac{3\sigma_1^4 - \left(\sigma_1^2\right)^2}{2\sigma_A^4} = \frac{\sigma_1^4}{\sigma_A^4} = \left(\frac{\sigma_1^2}{\sigma_A^2}\right)^2 = \left(\frac{\sigma_A^2\theta + \sigma_W^2}{\sigma_A^2}\right)^2$$

and the CRB can also be derived as follows:

$$I(\theta) = -E\left\{\frac{\partial^2 \ln L(\theta)}{\partial\theta^2}\right\} = -E\left\{\frac{\partial}{\partial\theta}\left[-\frac{\sigma_A^2}{\sigma_A^2\theta + \sigma_W^2} + \frac{\sigma_A^2\left(x_1^2 + x_2^2\right)}{2\left(\sigma_A^2\theta + \sigma_W^2\right)^2}\right]\right\}$$
$$= -E\left\{\frac{\sigma_A^4}{\left(\sigma_A^2\theta + \sigma_W^2\right)^2} - \frac{\sigma_A^4\left(x_1^2 + x_2^2\right)}{\left(\sigma_A^2\theta + \sigma_W^2\right)^3}\right\} = -\frac{\sigma_A^4}{\left(\sigma_A^2\theta + \sigma_W^2\right)^2} + \frac{\sigma_A^4 E\left\{x_1^2 + x_2^2\right\}}{\left(\sigma_A^2\theta + \sigma_W^2\right)^3}$$
$$= \frac{2\sigma_A^4\left(\sigma_A^2\theta + \sigma_W^2\right)}{\left(\sigma_A^2\theta + \sigma_W^2\right)^3} - \frac{\sigma_A^4}{\left(\sigma_A^2\theta + \sigma_W^2\right)^2} = \frac{\sigma_A^4}{\left(\sigma_A^2\theta + \sigma_W^2\right)^2} = \left(\frac{\sigma_A^2}{\sigma_A^2\theta + \sigma_W^2}\right)^2$$

As a consequence:

$$CRB(\theta) = I^{-1}(\theta) = \left(\frac{\sigma_A^2\theta + \sigma_W^2}{\sigma_A^2}\right)^2 = MSE\left\{\hat{\theta}_{ML}\right\}$$